WeBank Technology Services

WeBank Tokenization platform

March 2025

Strategic Engagement in Web3 to Facilitate and Empower Business

1 — FISCO 金链盟

- FISCO Blockchain organization Initiation

2 — FISCO BCOS

- Open-source infrastructure

3 — WeBank Web3 Ecosystem

- Web3 Ecosystem in WeBank

4 — HKChain

- An infrastructure rooted in HK and oriented towards the globe

…… — More to expect

Blockchain Tech-stack for Banking

e.g. Provide technology infrastructure for digital banks' digital assets business

Web3.0 Infrastructure Construction & Operation

e.g. Support the Web 3.0 ecosystem in HK as an infrastructure provider

Blockchain-based Business Scenarios Implementations

e.g. RWA for various industries, virtual assets, TRA etc…

FISCO - One of the Largest Blockchain Organization in China

Financial Services Blockchain Consortium (Shenzhen)

- Nonprofit organization established in May 2016
- Founded by financial institutions & tech companies
- Over 100 members

Number of Members: 100+

500+
Benchmark Applications

100+
Consortium Members

5000+
Institutional Participants

100K+
Community Members

Banks

WeBank 微众银行 · 中国农业发展银行 Agricultural Development Bank of China · 中信银行(国际) China CITIC Bank International · 厦門國際銀行 Xiamen International Bank · 中原银行 Zhongyuan Bank · 华润银行

Insurance companies

太平洋保险 CPIC · 前海人寿 Foresea Life Insurance · 富德保险控股 Funde Insurance Holding · 泰康 Taikang · 华安保险 Sinosafe Insurance · 中保车服

Securities companies and asset management firms

CMS 招商证券 · 国信证券 Guosen Securities · 广发证券 GF Securities · 长城证券 Great Wall Securities · 第一创业 First Capital · 博时基金 Bosera Funds

Other well-known institutions

ICT · China Tower 中国铁塔 · CEC 中国电子 · 数字广东 Digital Guangdong · Tencent · 科大讯飞 iFLYTEK · SF Digits 顺丰数科

FISCO BCOS - Underpinning Key Digital Infrastructure Nation & Industry Wide

FISCO BCOS

- **2015**
  In-house R&D of consortium chain platform
- **2016**
  FISCO (NPO) founded
- **2017** Open source
  FISCO BCOS v1.0
- **2019**
  FISCO BCOS v2.0
- **2021**
  FISCO BCOS v3.0
- **2022**
  FISCO BCOS v3.1
  - 1k+ full nodes with >100k TPS
  - 300+ applications
- **2023**
  Research paper accepted by Supercomputing Conference 2023

Widely Acknowledged Open Source Platform

Guangdong-Macao Cross-border Data Verification Platform

- Officially launched in April 2022 to facilitate the swift and orderly flow of cross-border data between Guangdong and Macao
- Cited on the official website of the Central Government of China

Shenzhen-Hong Kong Cross-border Data Verification Platform

- The case was selected as one of the “Ten Major Events in Shenzhen Finance in 2023” by Finance of Shenzhen Municipality

ESGC Benchmarking Report Application

- First Official Organization to Introduce FISCO BCOS Blockchain to ESG Regulatory Efforts
- Helping SMEs to better measure industry ESG performance

*As of January 2024

A Robust & Secure Platform for Financial Industry Needs

>200k	>1k	180+	100%
TPS	Institutional full nodes capacity	Source code repositories	Tested by 3rd party certification body

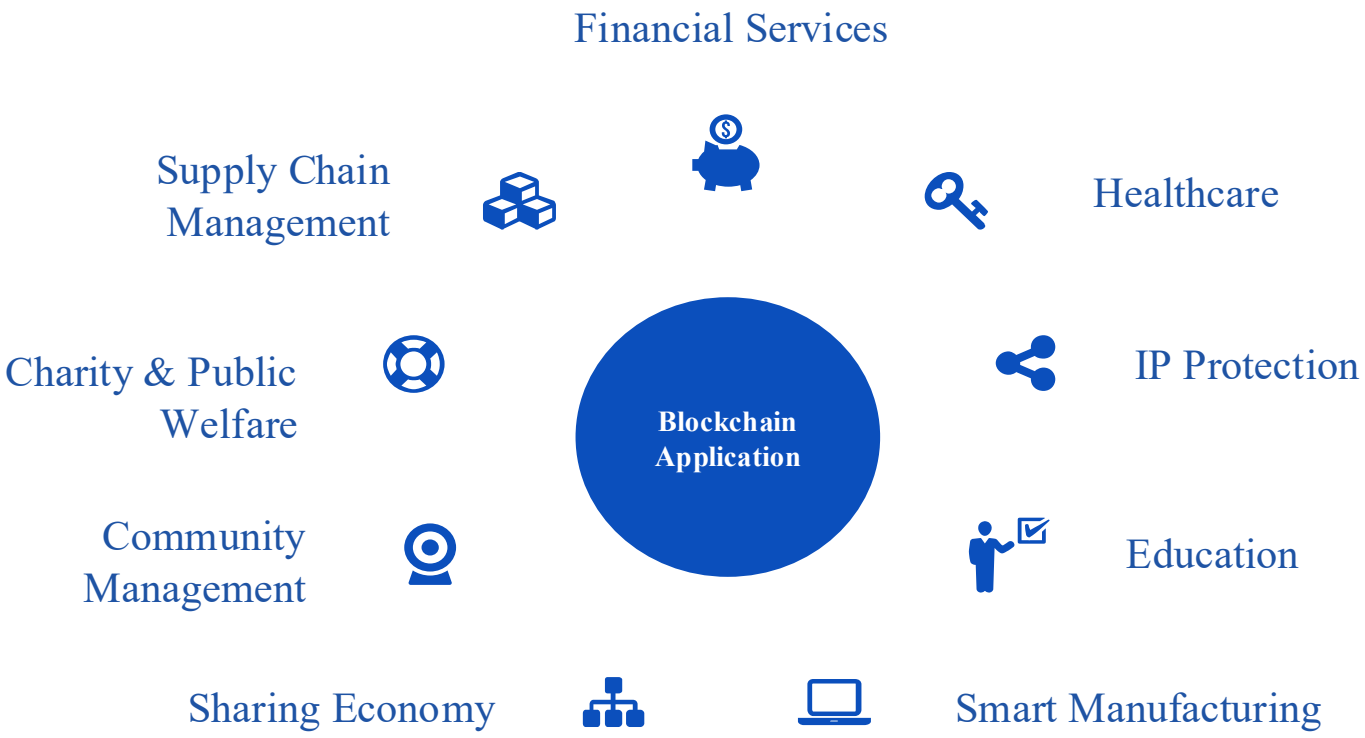
High Performance | Reg-tech Ready | High Security Standards

Type	Feature
Performance	• TPS > **200K**[1] • Latency < **50ms** • Finality= **1 Block**
Scalability	• RPBFT Consensus Algorithm, supporting **thousands** of consensus nodes. • **Distributed network** that supports multiple node types, multiple roles, and various applications.
High Availability	• Financial-grade high-availability architecture supporting **multi-region** **active-active** deployment. • **Observable architecture** supporting monitoring scripts, metrics reporting, and visual monitoring.
Governance	• Supports **RBAC (Role-Based Access Control)** permissions without relying on smart contracts for implementation. • On-chain functions are **customizable** and support **online upgrades**. • Supports regulatory browsers, on-chain auditing, and on-chain voting for on-chain governance. • Supports **Web3 DAO (Decentralized Autonomous Organization)** governance solutions.
Security	Multiple ledger architectures to accommodate different scenarios (1:1, 1:n private transactions)： • **Multigroup ledger** based on network isolation, offering high throughput. • **Cryptographic ledger** based on ZK (Zero-Knowledge) technology, providing strong privacy.

Empower Business in Various Scenarios & Applications

5000+
Institutional Participants

100K+
Community Members



Expand Business Scenarios

Explore business scenarios with partners

- Virtual assets
- RWA
- Digital ecosystem
- More…

Invite Trusted Partners

A trusted partner is invited as a node validator to jointly maintain the stability and security of the HK Chain

Build up Infrastructure

Build Layer 1 blockchain, use blockchain capabilities and technologies to solve problems within the ecosystem

Feel free to contact us for collaboration: kxzhang@webank.com

Awards & Recognitions

Local awards

2018
First Prize
Shenzhen Fintech Special Award

2020
Excellent Open Source Projects in China

2022
Third Prize
Financial Technology Development Award

2024
First prize
CCF (China Computer Federation)
Technology Invention Award

International awards

2019
Best Blockchain Project

2022
Forbes magazine unveiled its 2022 Global Blockchain 50 list.
Forbes magazine unveiled its 2023 Global Blockchain 50 list.

2022
Among international standardization achievements in blockchain published by ISO, 2 out of 4 Chinese use cases based on FISCO BCOS

2023
《SC23：FISCO-BCOS: An Enterprise-grade Permissioned Blockchain System with High-performance》

WeBank Blockchain - New unified public infrastructure for Web3 ecosystem

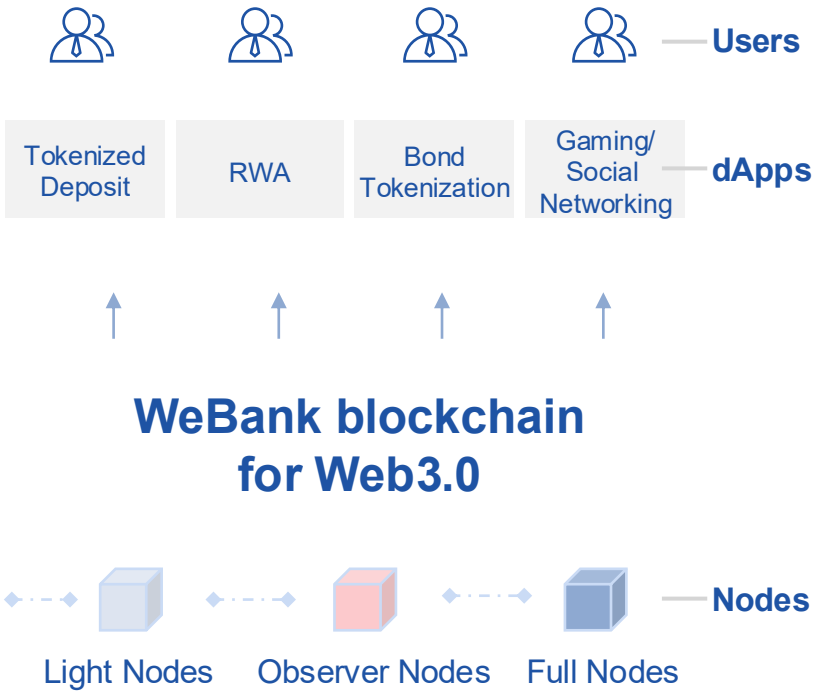
Why

- Existing solutions are flawed in respects of territorial regulation, general accessibility, etc. A new type of public infrastructure is a prerequisite to support a sustainable Web3.0 ecosystem.
- Fragmented infrastructures hinder the ecosystem development across regions and industries. A unified infrastructure is needed to support application development and integration.

What

- The WeBank blockchain takes into account the five pillars of a trusted infrastructure to support the sustainable development of digital economy and is a feasible path to build a new, unified public infrastructure for Web3.0 ecosystem.

How



Performance Overview

≈500m
Accum. txns

≈400k
Avg. daily txns

50+
dapps

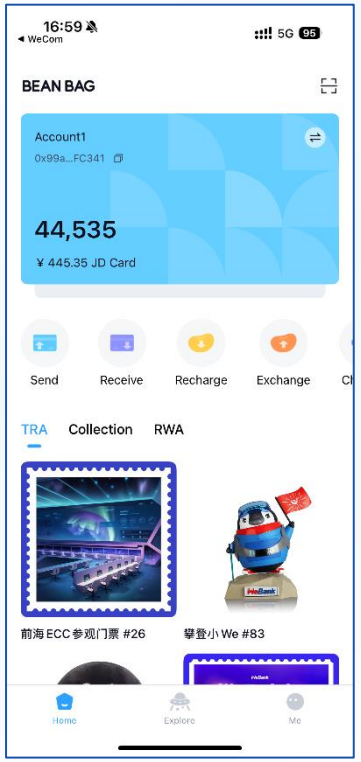
<$15k
Monthly operation cost (single chain)

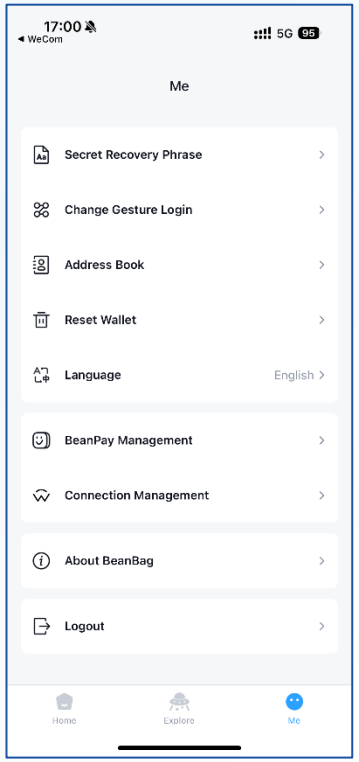
Personal Wallet: BeanBag

BeanBag: BeanBag supports functionalities including Recharge, Exchange, transfers, and transaction record queries.

Basic Functions

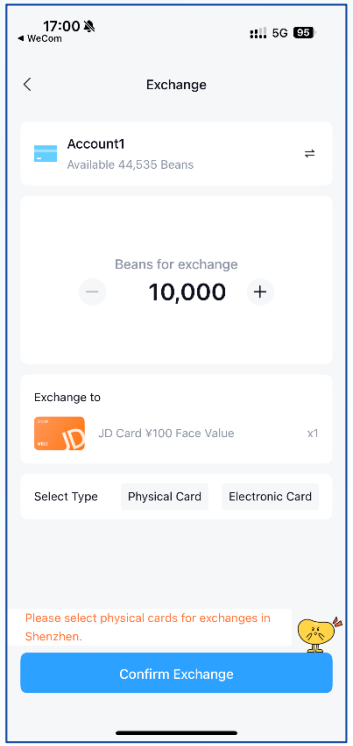
- 【Home】Account Overview
- 【Me】Wallet Management





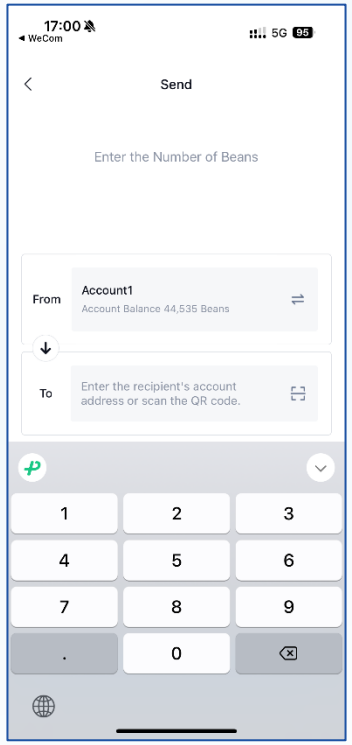
Recharge & Exchange

- 【Recharge】Fiat Currency to YuanDou Conversion
- 【Exchange】YuanDou to JD Card Exchange



Transfer & Transaction

- 【Send】YuanDou Transfer to External Addresses
- 【History】YuanDou Transaction Records Query

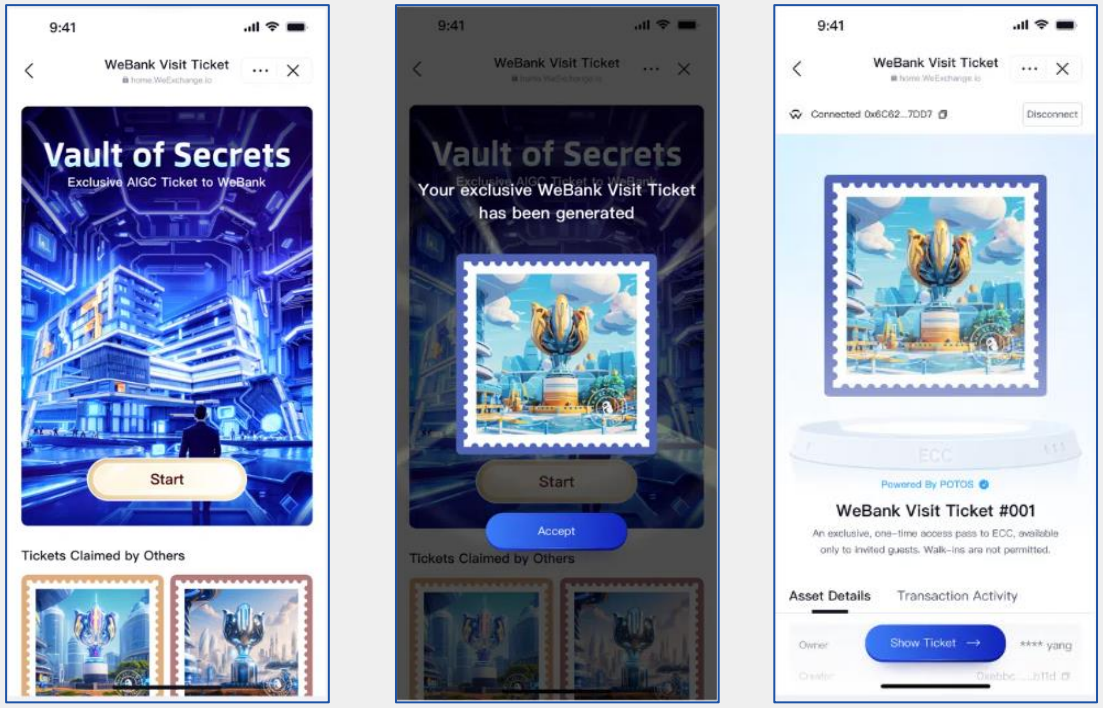


Weco Ecosystem in Hong Kong: Exploratory Implementation Usecase

HK FinTech Week—Admission Ticket

01

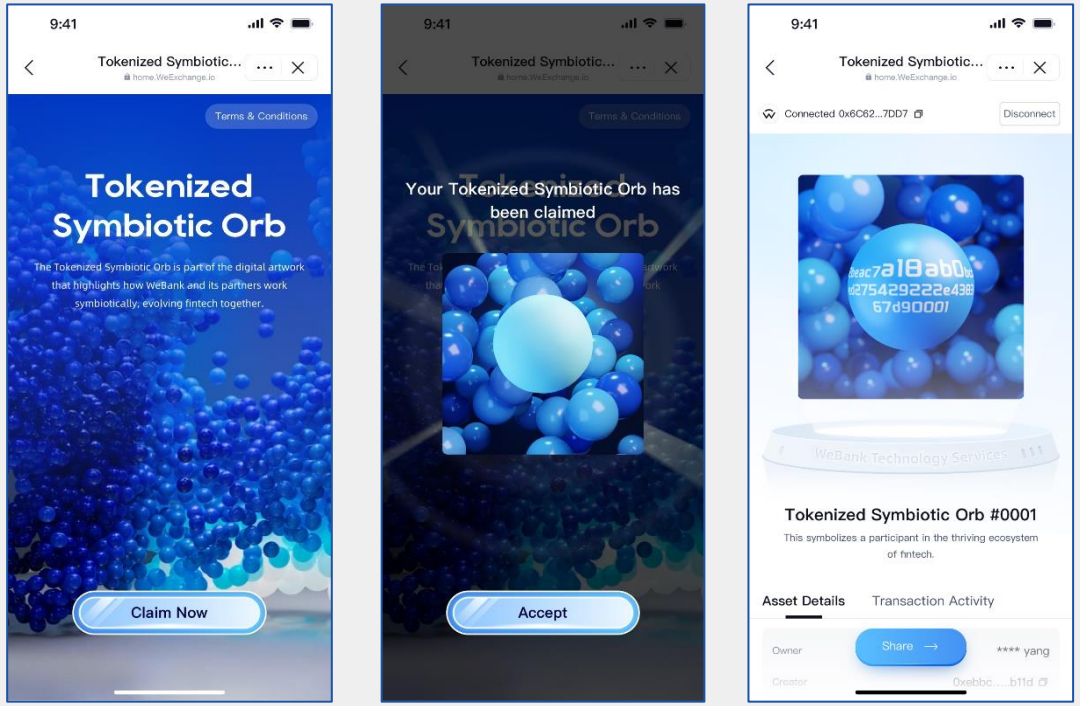
Admission Ticket for Special Guest



Wetech(Hongkong) Opening Ceremony

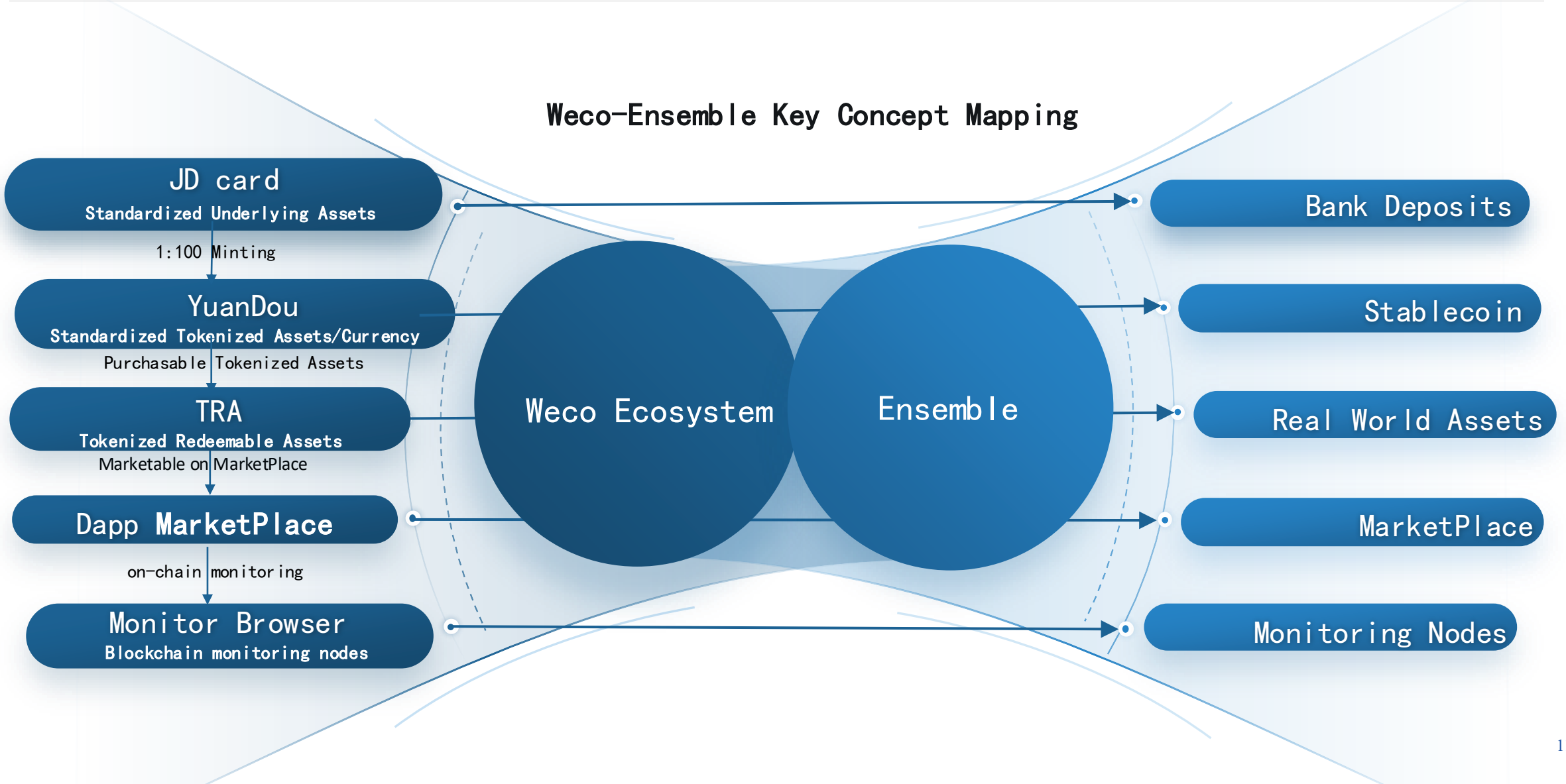
02

Voucher for Special Guest



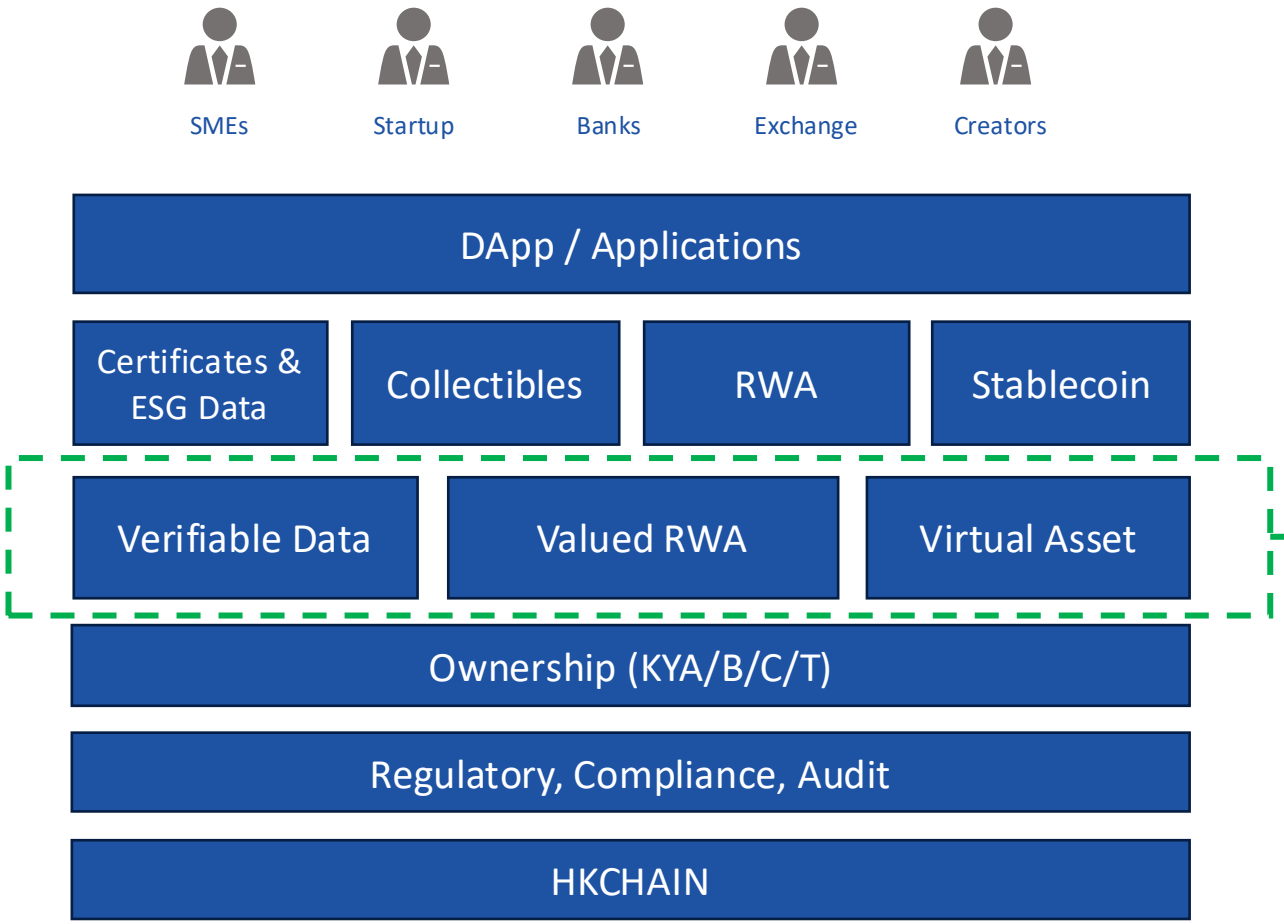
Weco-Ensemble Sandbox Scenario Interoperability

Weco Sandbox has end-to-end validated the complete process of digital asset transactions, demonstrating interoperability with the Ensemble Sandbox scenario.



4. HKCHAIN – Strategic Goals for Future Growth

Creating a secure, compliant, and Hong Kong-based Web3 blockchain ecosystem



Enabler

1. **Verifiable Data**
   Focus on securing and verifying data integrity during transfer, enabling businesses to validate data through golden source on HKCHAIN, enhancing trust and operational security.

   Supporting Scenarios: ESG Data, Assessments, Reports, Certificates, Awards

2. **Valued RWA**
   Tokenization of cash-flow-generating RWAs, allowing businesses to enhance transparency and efficiency in asset management on-chain.

   Supporting Scenarios: Intangible Assets (IPs), Equipment (GPUs), Financial Products (Bonds)

3. **Virtual Asset**
   Partner with Fiat-Referenced Stablecoin (FRS) Issuers to enable DvP on chain.

   Supporting Scenarios: Payment/Transfer, Cross Border Payment/Transfer, FX related business

Path to Digitalization

Tokenization Platform Overview

Phase 1

Phase 2

Tokenization Platform Management

Roles

- Issuer
- Investor
- Operator
- Custodian
- Issuer Agent
- ...

Asset Type

- Zero-Coupon
- Fixed-Rate Bonds
- Floating-Rate Bonds
- Other Bonds / Assets

Components

Wallet Connect
- Users connect to the platform via web3 wallet connect for sign up / login.

Issuance
- Tokenize the asset they plan to issue
- Supporting Zero or fixed rate bonds

Subscription
- Investors subscribe to listed asset tokens that using accepted units

Allocation
- Issuer sets allocation rules for investors after the subscription period ends.

Coupon
- Issuer provides the coupon payments based on the agreed terms
- Investor needs to accept the coupon payments

Redemption
- Token can be redeemed on the maturity date for the purchased unit

KYA
- Supports **W3C DID** for secure asset identification. Custodians can upload proof of holding the underlying asset.

...

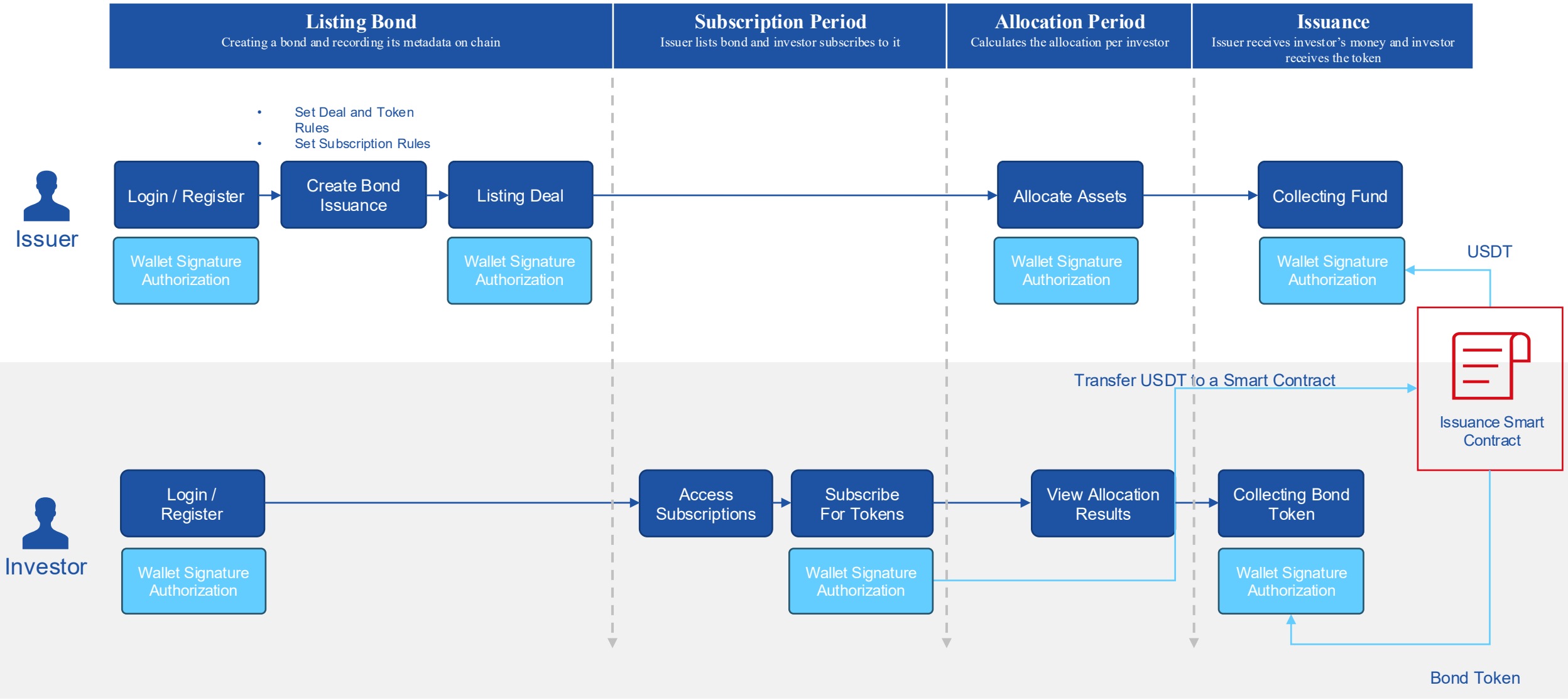
Smart Contract

Blockchain

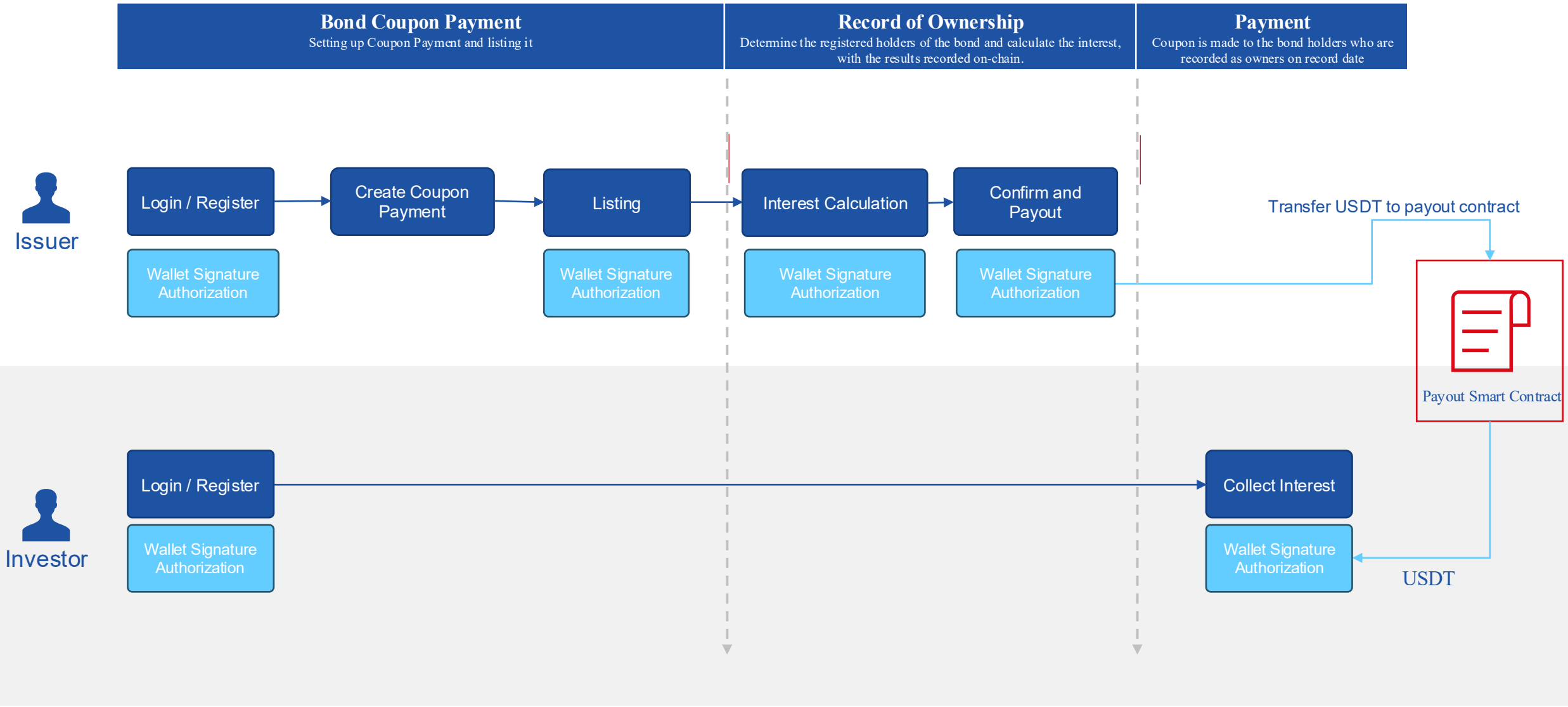
Tokenization Platform Feature Overview

Feature	Note
Account Setup	Member Creation ✓ Member sub-account creation ✓ Permission management ✓
Settlement Token mint and burn Issuance	Token mint creation ✓ Token burn ✓ Real time mint/burn of Settlement Token for on/off ramp operation or bridges with other tokenised platform
Normal Bond Issuance	Bond token creation (may support 2 to 3 bond types) ✓ Approval process ✓ Subscription ✓ Consolidation ✓ Manual allocation support with file upload (nice to have) Distribution ✓
Secondary DVP & FOP	Input instruction Matching DVP flow (Bond token vs Settlement token) FOP flow Account transfer
Corporate Actions	Coupon payment calculation and distribution (Should we use settlement tokens for coupon payments? Yes) ✓
Redemption Flow	Calculate redemption amount owed to DP ✓ Settlement Token transfer ✓ Bond token redemption ✓
Reconciliation	Mechanism to reconcile the sum of Bond Token held by all accounts is equal to the total nominal amount of the issuance size
Interfaces	Integration layer

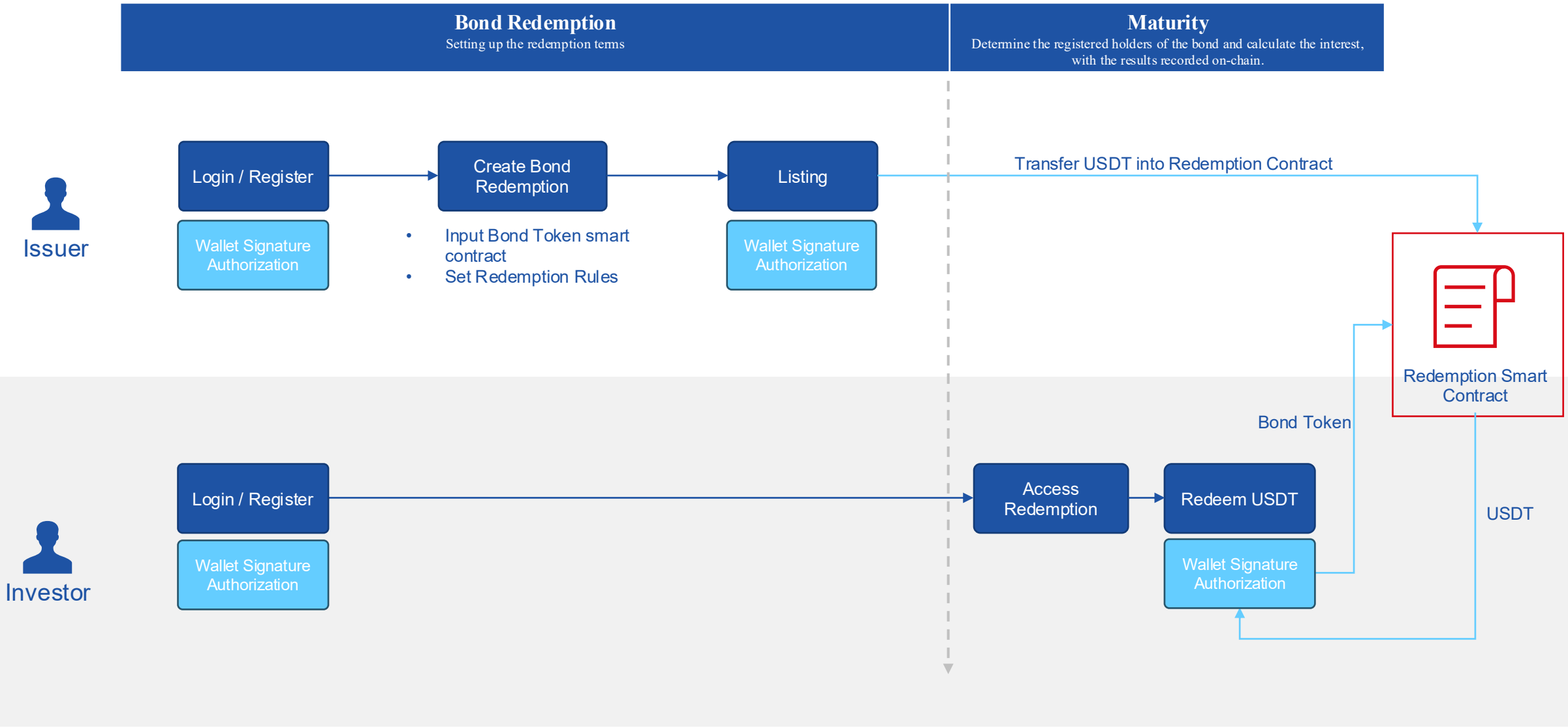
Issuance and Subscription Process



Coupon Distribution Process



Redemption Process



WeBank Technology Services

Thank You!

Tokenization Platform Feature Overview

Feature	Available	Note
Token Lifecycle management	✓	Support bond issuance, coupon, redemption on a blockchain.
Coupon calculation	✓	Coupon allocation and calculation automation
Audit Trail	✓	Application logs for audit purpose Support dual recording on both traditional databases and blockchain ledgers
KYB/KYC	✓	Support KYB/KYC for compliance and regulatory
ERC20 Token Standard	✓	ERC20 supports high interoperability and cross-chain transfers via bridges
Chain agnostic platform	✓	Optimized for FISCO BCOS but support any EVM-compatible blockchain
Secure Key Management	✓	Built-in key management system
Issuer/Investor dashboard	✓	Web-based dashboard UI for issuers/investors
Onchain authorisation	✓	Support secured authorisation via Web3 wallets
Investor subscription portal	✓	Web interface for browsing list of bond and subscription
Multi-language support	✓	Traditional Chinese and English
Data privacy	✓	Zero knowledge , MPC solution available for data privacy